

The Socioeconomic Effects of Forced Displacement: Evidence from the Tennessee Valley Authority*

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Abstract

We examine the socioeconomic effects of forced migration by focusing on individuals who were displaced by the Tennessee Valley Authority (TVA) dam projects in the 1930s. We use data from the relocation program associated with the TVA and link them to US Census data. We compare individuals who were impacted by the dam-induced flooding to those who will be displaced by different TVA dam projects at a later date. We find evidence that individuals who were displaced by the dam projects are more likely to participate in the labor force, marked by an inflow of labor into unskilled occupations and an outflow from skilled occupations. We examine racial disparities in outcomes and find that, after the relocation, Black men are more likely to be in these unskilled occupations compared to White men.

Keywords: Displacement, Labor, TVA

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1 Introduction

Forced displacement resulting from wars, natural disasters, and human rights violations has become increasingly common. The United Nations estimated the number of forced displaced individuals globally reached roughly 120 million at the end of 2025 [United Nations High Commissioner for Refugees \(2026\)](#). Understanding the consequences of forced displacement is therefore becoming increasingly important. Its severe impacts can likely persist long after the incident. Previous studies have shown evidence of negative impacts of relocation driven by natural disasters on labor market outcomes [Mueller and Quisumbing \(2009\)](#) and financial resources [Chen \(2020\)](#). Other studies have examined the impacts of the Great Migration on African Americans in the United States (U.S.) [Tolnay \(2003\)](#). There is little evidence on short- and long-term impacts of forced displacement induced by government development projects, although this practice of eminent domain is very common in the U.S.

This paper investigates the effects of forced migration by leveraging a natural experiment from the 1930's - the Tennessee Valley Authority (TVA) dam projects. The TVA created dams in seven states in the southeastern region of the U.S. as part of the New Deal, displacing thousands of people. We use data on the displaced individuals, which we have matched to U.S. Census records using a probabilistic matching algorithm, to examine the effects of forced displacement on a range of medium-run socioeconomic outcomes. Development of these reservoirs was an effort to combat the lack of economic opportunity in the region, especially after the downturn of the Great Depression. However, the success of this New Deal program on the people it displaced is not widely understood.

The Tennessee Valley Authority Act of 1933 marked the first time a government agency had power to govern the development of an entire region. Not only did the act improve infrastructure and rural life, it drastically improved the region's standard of living within a relatively brief period [Walker \(1998\)](#). "The T.V.A. has relocated over 12,000 farm families, moved entire villages, helped whole communities plan a better future, and withal it has demonstrated that social planning can win the enthusiastic support of the

people at the muzzle end of it. So the T.V.A. has set a pattern for social planning for a generation" Carr (1948). However, historical research suggests the TVA perpetuated segregation and discrimination against Black individuals Jones (2016); Alderman and Brown (2010). For instance, using TVA's relocation records and oral histories, Walker (1998) argues the Authority's benefits were not fairly distributed. The agency maintained a policy of institutionalized segregation and discrimination against Black people which also prevented them from deriving direct benefits from TVA's employment and educational programs. Walker (1998) also states that the beneficial programs provided by the TVA favored prosperous landowners over poorer landowners and tenants of both races. Although relocation workers attested to abiding by TVA's removal worker's code of ethics, many assumed African Americans did not have a sense of community, hence only using the word "community" in documentation towards White families. This suggests that TVA workers were less sympathetic about removing Black families from their communities than for White families. A chief criticism of the dam construction project is the lack of a systematic follow up of the removed families to their new locations Brown (1951). The author states, "If this was done, a better basis would be available for judging the success and failures of family removal and relocation in the Tennessee Valley."

This paper provides that missing systematic follow-up. Using a difference-in-differences design, we find that forced relocation by the TVA increased displaced men's labor force participation by 3.3 percentage points (10.2%), though displacement had no effect on the extensive margin of employment. Displaced men were also more likely to work in unskilled occupations and less likely to be in skilled occupations, a pattern driven primarily by an influx into non-farm unskilled roles rather than agriculture. These results suggest that while forced relocation may boost labor market entry, these gains are accompanied by a shift toward lower-skilled work rather than upward occupational mobility.

We further find differential impacts by race: Black men were 13.6% more likely than White men to be in the labor force and 53.8% more likely to be in unskilled occupations following displacement. Using a race-matched sample, we show that the contraction in skilled employment among displaced Black men was nearly triple the effect size seen

among White men, a disparity that is statistically significant and consistent with differential treatment in the post-displacement labor market. We also find that displaced male renters, rather than owners, drove the improvements in labor force participation and employment, while displaced owners instead saw declines in employment and skilled occupations. These patterns do not appear to be explained by selection into treatment or by relocation to systematically better counties, suggesting instead that they reflect the disruptive effects of displacement itself.

This study contributes to different areas of the literature. First, it adds to the body of work that examines the effects of displacement through the lens of the Great Migration in the United States. Although the identification strategy in these studies is affected by the fact that African Americans made a decision to migrate north for better opportunities, harsh living conditions have been argued to leave many with little choice but to relocate. Existing evidence shows that relocation had positive impacts: increased earnings of Black migrants [Collins and Wanamaker \(2014\)](#) and schooling for migrant children, which decreased the White-Black education gap [Baran et al. \(2024\)](#). However, [Dernoncourt \(2022\)](#) finds that destination cities responded to an influx of Black migrants by changing city characteristics, such as increasing policing, thus reducing the benefits of migration. This response to an influx of Black migrants also led to an increase in incarceration rates for Black individuals [Eriksson \(2019\)](#). We add to these studies by examining how Black and White people respond similarly and differently to a more short-distance forced relocation.

We also contribute to the literature on the impact of displacement after natural disasters. For example, previous studies have shown that Hurricane Katrina had small impacts on victims' wage earnings and income, with labor supply and earnings returning to pre-disaster levels for those who relocated back after the hurricane [Deryugina et al. \(2018\)](#); [Vigdor \(2007\)](#). However, these natural disasters can cause financial hardship, especially for poorer countries and counties with greater income disparity [Johar et al. \(2022\)](#); [Cavallo et al. \(2022\)](#); [Lim et al. \(2017\)](#). The [National Research Council \(2006\)](#) states: "...more recent research suggests that such [positive rebounding from natural disasters] findings paint an overly simplified and perhaps overly optimistic picture of post-disaster recovery."

In addition, we contribute to the literature on the impact of displacement caused by government development programs. For instance, [Chyn \(2018\)](#) and [Chetty et al. \(2016\)](#) examine the impact of distinct policies within the Moving to Opportunity literature that led to the demolition of public housing and provided housing vouchers to move people from high-to-low-poverty neighborhoods in the 1990s. They show that this form of displacement improves the economic outcomes of children who moved. These studies raise the question of whether providing an opportunity to move, rather than forcing relocation, is better for citizens in that area. The results of [Chyn \(2018\)](#) suggest that forcing relocation leads to beneficial longer-term outcomes for children, perhaps because short-term constraints prevented such moves when they were not forced. However, development projects from the 1996 Atlanta and 2008 Beijing Olympic games pushed migrants further out into poorer areas, resulting in negative long-term impacts [Richardson \(2022\)](#); [Shin and Li \(2013\)](#). The literature on the interstate highway system has documented its negative impacts, especially on the African American community, by destroying communities and creating inner-city ghettos [Karas \(2015\)](#). However, in India, the creation and expansion of highways improved the socioeconomic well-being of people living in its proximity (although they were not displaced) [Sengupta et al. \(2007\)](#).

We add to the ‘New Deal’ literature by examining the effects of its establishment. Many New Deal programs have been shown to compensate the repercussions of the Great Depression [Fishback et al. \(2007\)](#). The authors saw that an increase in relief spending from these New Deal programs led to a decline in infant mortality, suicide rates, and deaths from infectious diseases. The majority of grant and loan programs were geared towards the poor and unemployed, raising income and increasing property values near new public housing projects, while also reducing property crime rates [Fishback and Kachanovskaya \(2016\)](#); [Fishback and Kollmann \(2015\)](#); [Fishback et al. \(2010\)](#). However, certain programs also led to a reduction in employment and labor force participation, especially for women [Blundell et al. \(2016\)](#); [Sundstrom \(2001\)](#).

The existing economic literature on the TVA has not been able to examine individual responses to the program. The TVA reservoirs led to a temporary increase in malaria

rates, later reduced by TVA’s anti-malaria programs [Kitchens \(2013\)](#). Its electrification had no significant effect on the productivity of farming or manufacturing [Kitchens \(2014\)](#). However, [Kline and Moretti \(2014\)](#) find that the growth rate of manufacturing employment was significantly larger in TVA counties than non-TVA counties; hence, TVA sped the industrialization of the Tennessee Valley. The authors focus on the economy of TVA counties compared to non-TVA counties, while our paper will focus on the people who were impacted by the development from the TVA.

This project contributes to the literature on forced relocation more broadly. Similar papers in this literature look at the impacts of large dam projects on local economic outcomes. Dams constructed in India and Africa had negative impacts, such as increased poverty and lower productivity croplands, on those near the construction versus those living downstream of the dam, who benefited [Duflo and Pande \(2007\)](#); [Strobl and Strobl \(2011\)](#). Other papers focus on government administered forced relocation programs in Europe following war [Hornung \(2014\)](#); [Becker et al. \(2020\)](#); [Sarvimäki et al. \(2022\)](#), and catastrophic natural disasters [Nakamura et al. \(2022\)](#). Importantly, these migrations featured racially and ethnically homogeneous groups. A recent review of the forced migration literature notes that little work examines how “contextual variables” like race influence economic responses to forced migration [Becker and Ferrara \(2019\)](#). In our setting, we examine racial heterogeneity especially as the TVA relocation program occurred during the era of Jim Crow in the southern United States. We provide the first causal evidence of the effects of government sponsored forced relocation on individual labor market outcomes, and document disparate impacts for Black and White individuals.

Overall, our findings show that forced relocation from the Tennessee Valley Authority’s dam construction project improved labor market outcomes. The defining aspect of this project is that these events, like extreme weather events, are forced and citizens do not have a choice of whether to stay or leave their homes.

The paper is organized as follows. Section 2 provides background information on the creation of the Tennessee Valley Authority and its relocation program. Section 3 describes the data sources used to identify forced relocation effects; it also contains an explana-

tion of our linking process to historical census. Our empirical approach is discussed in Section 4, and includes our conceptual framework. The results are presented in Section 5 along with a robustness check and a discussion of mechanisms. We conclude in Section 6.

2 Background

2.1 Historical Background: Tennessee Valley Authority and the Great Depression

The 1930s Great Depression was one of the worst economic depressions in history. During the time, many Americans placed blame on President Herbert Hoover for steep unemployment and massive bank failures. His successor, President Franklin D. Roosevelt, began the ‘New Deal’ recovery program to stimulate economic recovery. The Tennessee Valley Authority (TVA) was established by the Tennessee Valley Authority Act of 1933 as part of the New Deal. The duties of the organization included improving the navigability of the Tennessee River; providing flood control through reforestation of marginal lands in the Tennessee Valley watershed; developing agriculture, commerce and industry in the valley; and operating the hydroelectric dams¹. The Act also gave TVA the power of eminent domain, allowing them to acquire land along the Tennessee River for developmental reasons. These hydroelectric dams, shown in Figure 1, were constructed to expand the Tennessee River, moving east to west, and runs through the seven states that TVA oversees: Alabama, Georgia, Kentucky, Mississippi, North Carolina, Tennessee, and Virginia. Construction of these dams caused numerous towns to be submerged underwater, forcing residents to relocate. On average, construction of each dam took about 3 years to complete between 1933 and 1953.

2.2 Tennessee Valley Authority’s Relocation Program

The TVA provided a relocation program to assist families displaced by dams move to new areas. TVA sent interviewers to each household that would be affected. These interviewers filled out a relocation application with the family that contained detailed information about their house, land, and demographics for all individuals living there.

¹Transcript of the [National Archives and Records Administration](#) (1933)

The TVA used this information to determine suitable housing for families to relocate. Although TVA did not provide compensation for relocation, it provided its resources to find new homes for sale, appraised them, and relocated families to farms that best suited their needs [Satterfield \(1937\)](#). To our knowledge, no rubric exists that details how and in what order new dwellings were chosen. However, it has been documented that families were relocated to a new home and land that was similar to their previous home and land [McCarthy \(1949\)](#); [Satterfield \(1937\)](#). Although most families were tenant farmers, families and landlords who owned their property were offered fair market prices [McCarthy \(1949\)](#). Relocation was provided for both the living and deceased², and an estimated 15,000 families and 30,000 graves were relocated across all seven states.

To our knowledge, there were no clear documented rules determining the TVA's decision to flood particular areas. However, [Howard \(1936\)](#) explains the steps towards dam construction the TVA took in order to implement the program as efficiently as possible. In the twelve step plan towards the completion of the project, fully interpreted as a list in the Appendix Figure [A1](#), the majority of the steps focused on the employees hired to construct the dams. These included understanding the labor supply of the region, as well as providing resources for those employees such as a camp to house them and resources to educate them on-and off-duty. Other steps focused on cooperation between local governments and the TVA as the region developed and the project progressed. One of the last steps was the relocation of families. As mentioned, detailed facts about families who were in the path of the dam construction were obtained; these details were used by social scientists to analyze those who were being displaced and to determine the relocation strategy. [Howard \(1936\)](#) also notes that an analysis determining the extent to which [relocated] families have successfully integrated themselves into a new life, without either cultural or economic loss, will be important to document the success of the program. To our knowledge, a full analysis as such has yet to happen³.

²The Tennessee Valley Authority obliged by the family's wishes to relocate their loved one's grave.

³TVA issued a descriptive report on the resettlement of the population of each major dam constructed in the 1930s. We will provide causal inference about resettlement using economic models.

3 Data

3.1 Information on Relocated Families

We obtained information on relocated families in Alabama and Tennessee from the “U.S., Tennessee Valley, Family Removal and Population Readjustment Case Files, 1934-1953” dataset via ancestry.com. This dataset contains PDF files of original relocation applications that provide rich information about each relocated family. Figure 2 shows a sample application that displays housing details such as property ownership, property characteristics, and crop farming style described at the household level. Employment information such as occupation, income, and the highest level of education was collected for all family members who currently reside in the house. For each resident, demographics such as gender, race, birth year, relation to the head of the household, along with their full name were provided as well. TVA utilized these applications to help with moving decisions. To our knowledge, we are the first to transcribe and digitize PDF files of TVA’s relocation applications for the states of Alabama, Tennessee, and Georgia, which constitutes more than 7,800 relocated families.

3.2 Census Data

After obtaining information on which families were relocated, we linked those families to the 1930 census by matching the name and other defining characteristics such as birth year, sex, and residing county to gather our outcome variables. The census also allowed us to gather outcome variables for non-relocated families who were not yet in the path of the dam construction, which constitutes our control group. These include employment status, participation in the labor force, and occupational classification codes. We obtain data prior to dam construction from the 1930 census, with post-construction data coming from the 1940 census.

We use record linking methods to link the data on relocated families to the 1930 census data using a probabilistic matching algorithm that matches individuals on name, birth year, location, and gender. We require an exact match on county and gender, and

allow for flexible matching on both name and age. However, we require that individuals be born in the same decade and have the same first and last initial. We are able to link 77% of the relocated families data to the 1930 census. Of the relocated individuals linked to 1930, 81% were also linked to the 1940 census. The population of linked relocated families contains more males, White individuals, farmers, have higher income and literacy rate on average, and less likely to rent their property compared to those who did not link across earlier censuses.⁴ Our final sample consists of individuals in the state of Alabama, Tennessee, and Georgia who have census information from 1930 and 1940.

4 Empirical Approach

4.1 Empirical Strategy

To understand impacts of a forced relocation, we compute a simple difference-in-difference model that observes outcomes of those who were forced to relocate, the treated group, compared to those who were similar but not-yet forced to relocate due to timing of the dams construction, the control group, after the TVA Act was enacted in 1933. These individual labor outcomes consist of employment status, labor force status, and the switch to unskilled or skilled occupations [Ferrie \(1997\)](#). We look solely at men because married women were not consistently asked about employment when censuses were conducted. Therefore, women and children are excluded from individual employment outcomes.

Specifically, our treatment group contains individuals displaced by the Wheeler (1933) and Guntersville (1935) dams in Alabama and the Norris dam (1933) in Tennessee. The control group contains individuals displaced by the Cherokee (*late 1940*), Fort Loudoun (*late 1940*), Ocoee No. 3 (1941), Douglas (1942), Boone (1950) and Fort Patrick (1951) dams in Tennessee and the Nottely dam (1941) in Georgia. The placement of the dams and the locations of individuals in our treated and control groups are shown in Figures [1](#) and [3](#) respectively.

⁴See Table [A2](#) for a detailed comparison of the linked and unlinked individuals

Equation (1) shows out difference-in-difference model:

$$Y_{it} = \beta_0 + \beta_1 Black_i + \beta_2 Treated_i + \beta_3 Post_t + \beta_4 Treated_i * Post_t + \tau_t + \epsilon_{it} \quad (1)$$

where Y_{it} represents the socioeconomic outcomes for an individual male i in census year $t \in \{1930, 1940\}$. $Black_i$ is a binary variable that equals 1 if the race of the male is Black and 0 if race is White⁵. $Treated_i$ indicates whether a male was displaced by the dam construction. $Post_t$ represents the decennial census period, 1940, after the relocation time period. The interaction between $Treated_i$ and $Post_t$ indicates a displaced male in the post period of 1940. β_4 is our coefficient of interest, representing the socioeconomic effects of being displaced by TVA's dam construction projects. Our error term is represented by ϵ_{it} . All regressions report county-clusted standard errors. For results by race, we use equation (2):

$$Y_{it} = \beta_0 + \beta_1 Treated_i + \beta_2 Post_t + \beta_3 Treated_i * Post_t + \tau_t + \epsilon_{it} \quad (2)$$

where we drop the $Black_i$ indicator variable and run the regression separately by race. β_3 becomes our coefficient of interest and the rest of the equation follows the description of equation (1).

4.2 Identifying Assumption

A key assumption underlying our difference-in-differences design is that treated and control counties would have followed similar trends in the absence of dam construction. Because our main analysis relies on decadal Census data, we cannot directly assess pre-treatment trends in the primary sample⁶. To provide indirect evidence for this assumption, we construct event study plots using six annual county-level proxy outcomes drawn from Fishback and Kantor (2018): population, CCC camp presence, total New Deal spending per capita, WPA and FERA spending per capita, manufacturing employment, and

⁵We drop nonblack and nonwhite races due to a small amount of observations. There is also no evidence of multiple races per household in our sample during this time period.

⁶For a detailed decadal summary of the statistics, please refer to the Appendix, Table A1

farming assistance.⁷ These outcomes span a range of demographic and economic dimensions that allow us to assess whether treated and control counties were trending similarly in the years leading up to dam construction. Importantly, many of these outcomes proxy for other New Deal programs that could potentially affect the labor market outcomes of individuals impacted by dam construction.

Figure 4 presents the results. Across all six proxy outcomes, pre-treatment coefficients are small, statistically indistinguishable from zero, and show no systematic directional drift in the years prior to dam construction. This pattern holds consistently throughout the pre-period before dam construction, suggesting that treated and control counties were on broadly similar trajectories before treatment. Although these proxies do not constitute a direct test of parallel trends for our primary outcomes, the consistency of the null pre-trend findings across six independently measured series provides reassuring evidence that our identifying assumption is plausible. Additionally, there does not appear to be a significant change in any of the proxy outcomes after dam construction, which provides suggestive evidence that we are capturing the true treatment effect of forced displacement.

4.3 Conceptual Framework

To inform our empirical work, we discuss the possible channels in which our outcomes may respond to forced relocation. First, consider the setting of our study; Alabama, Tennessee, and Georgia are primarily rural states that host many farmers on their land. With the creation of the dam project, the TVA sought workers from the local region to help with construction. As shown in the Appendix, Figure A1, the steps towards implementation of the dam construction focused mainly on the employees. With an emphasis on the employees and the job availability that the dam project would bring, we expect to see an increase in employment. As the availability for employment increases, a forced relocation might also provide the opportunity of occupation switching. For example, construction jobs for the dam project is seen as an unskilled occupation, hence we foresee an influx

⁷The CCC, WPA, and FERA are New Deal programs providing employment and monetary assistance to individuals. The CCC is the Civilian Conservation Corp, the WPA is the Works Progress Administration, and FERA is the Federal Emergency Relief Administration.

into unskilled occupations whereas we might see a decrease into skilled occupations.

With the Tennessee Valley Authority Act of 1933 occurring in the midst of the Segregation Era in the South, we expect to see differences in effects between Black and White relocatees. As stated above, we anticipate the dam construction project causing an increase in employment. However, this increase may have favored White men. This hypothesis is grounded in the historical documentation of the TVA, where seemingly neutral policies often led to unequal outcomes. [Walker \(1998\)](#) documents the “first come, first served” approach was frequently used to preserve community demographics, which systematically limited opportunities for Black families while favoring White families.

5 Results

5.1 Main Results

We report the effects of TVA relocation program on labor market outcomes in [Table 1](#) and graphically in [Figure 5 \(a\)](#). Our estimates indicate a 3.3 percentage point (p.p.) increase in labor force participation for displaced males compared to those who have not yet been displaced (a 10.2% increase). Although displacement had no effect on the extensive margin of employment, displaced men were 6.4 p.p. more likely to work in unskilled occupations and, suggestively, 4.2 p.p. less likely to be in skilled occupations.⁸ These results suggest that forced relocation increased labor market entry, but primarily through an expansion into unskilled, non-farm work rather than occupational upgrading.⁹

[Table 2](#), and [Figure 5 \(b\)](#) graphically, depicts differential impacts of forced relocation by race. Overall, both Black and White men see outcomes in the same direction as our main results, which indicates an increase in labor market participation and unskilled

⁸Unskilled occupations and skilled occupations are defined using the occupation classification scheme in [Long and Ferrie \(2013\)](#), where unskilled occupations include laborers and service workers and skilled occupations include managers, craftsmen, and other professionals. Our placebo test in [Table 3](#) reveals a significant pre-existing divergence in skilled occupation between 1920 and 1930, prior to dam construction. Given this evidence of a pre-trend, we do not interpret the skilled occupation result as a clean causal estimate. We report it for completeness, but our preferred estimates of occupational effects rely on the unskilled occupation result (column 3), for which the placebo test shows no evidence of differential pre-trends.

⁹We decompose the unskilled occupation category into non-farm and farm-related roles—following the classification in [Long and Ferrie \(2013\)](#)—to determine if the shift into unskilled work is driven by a movement away from agriculture. As shown in [Table A3](#), the increase in unskilled occupations is primarily driven by an influx into non-farming roles (Column 2), compared to the small incline in unskilled farm labor (Column 3).

occupations, and a decrease in skilled occupations. We run a triple interaction analysis using an indicator variable that helps identify any difference in outcomes by race. In the last row in Table 2, we report no significant difference in employment levels nor in skilled occupations for both Black and White men. This means that men, regardless of race, experienced similar treatment in these areas after relocating. However, we do show a significant difference in labor force participation and unskilled occupations. Specifically, Black men were 13.6% more likely to be in the labor force, yet 53.8% more likely to be in unskilled occupations compared White men. These results suggest differences in the impact of a forced relocation by the TVA between Black and White men.

5.2 Robustness Checks

We assess the validity of our identification strategy through a placebo test, a test of selection into treatment, and estimation on alternative matched samples. Table 3 reports a placebo test comparing 1920 and 1930 outcomes, a period entirely before dam construction authorized by the TVA Act of 1933. Consistent with parallel trends, we find no significant divergence in employment rate, labor force participation, or unskilled occupation share between eventually-treated and control counties.¹⁰ We do, however, find a significant pre-existing divergence in skilled occupation share of -11.8 p.p., which we address directly in a footnote of our main results discussion above.

Table 4 regresses treatment status on each of our four outcome variables measured in 1930, controlling for race, to assess whether observable labor market characteristics in 1930 predict subsequent selection into treatment. We find no evidence that any outcome significantly predicts treatment assignment, supporting the assumption that treatment status is not driven by pre-existing differences in observable labor market characteristics.

We further estimate our main specification on a county-matched sample (Table 5), both including all counties (Panel A) and separately excluding counties where dam construction occurred in the 1930s (Panel B) or 1940s (Panel C), to address concerns that our results are driven by any particular decade of dam construction. Our core labor force

¹⁰Coefficients range from -0.001 to 0.022 p.p. and are not statistically distinguishable from zero across these three outcomes.

participation and occupational results are stable across all three panels: displaced men remain 2.3-2.8 p.p. more likely to participate in the labor force, 5.5–5.8 p.p. more likely to work in unskilled occupations, and 2.2–3.4 p.p. less likely to work in skilled occupations, closely tracking our main estimates. Employment rate, insignificant in our main specification, becomes marginally significant and positive in the matched sample, suggesting our main estimate is if anything conservative.

Finally, because dam construction for all treated counties occurs within a single intercensus interval, our design is a standard two-period difference-in-differences estimator. As such, our estimates are not subject to the recently-documented biases arising from staggered treatment timing (e.g., [Goodman-Bacon \(2021\)](#); [Callaway and Sant’Anna \(2021\)](#); [Sun and Abraham \(2021\)](#)), since there is no variation in treatment timing within our estimation sample. Taken together, these results indicate that our findings are robust to alternative sample restrictions and are not subject to the biases that typically arise from staggered treatment adoption, with the exception of the skilled occupation result noted above.

5.3 Mechanisms

One potential mechanism driving the results is that individuals were relocated to places that were better at baseline before the TVA projects began. To examine whether this is the case, we calculated the average county level characteristics for each county in Alabama in the 1930 Census. We then assign the average county characteristics to each individual in our data based on their county of residence in 1930 and 1940, and estimate the difference-in-differences in equation (1). The results of this analysis are shown in Table 6. From this analysis, our small and insignificant results show that people displaced by the TVA projects moved to places that were largely similar to where they were living before displacement.

A second potential mechanism is that Black individuals who were relocated faced discrimination. To test this, we match the individuals in our treatment group with the same-race control individual who is the nearest neighbor on baseline demographics and

labor market outcomes. Then, we estimate the effect of forced relocation on this matched sample. Table 7 depicts the differential impacts of forced relocation by race using our tightly controlled, race-matched sample. Overall, matched Black and White men see insignificant and small affects on employment, labor force participation, and unskilled occupational sorting. Although these estimates move in the same direction as our main result findings, there is no significant difference in estimates by race. However, it is important to note that while insignificant, Black men see a 3.0 p.p. increase in unskilled occupations compared to a flat response of 0.3 p.p. increase for White men. Race-matched men experience a statistically significant reduction in skilled employment; this constitutes the sole statistical disparity, with the rate for Black males being nearly triple that of White males. This divergence provides evidence of potential differential treatment and racial discrimination within the post-displacement labor market, as the disruptive shock of forced relocation disproportionately penalized Black workers by restricting their access to skilled roles.

A third potential mechanism driving the results is that there is differential treatment between renters and homeowners. One potential concern related to this is that individuals were selected into treatment based on their household wealth or other baseline characteristics. We test this in Table 4 by regressing treatment status on our baseline outcomes in the 1930 Census. We find no evidence that individuals were selected into treatment based on observable characteristics. In Table 8 we examine whether there are differential outcomes between renters and owners. We find significant positive effects on labor force participation and unskilled occupations, and a significant negative effect on skilled occupations for displaced male renters compared to not yet displaced male renters. We see no effects in labor force participation and unskilled occupations for displaced men who were owners. However, there is a significant decline in the employment rate and skilled occupations for displaced male owners compared to their not yet displaced counterparts. The last row in Table 8 show these estimates are significantly different for renters and owners. Specifically, displaced male renters are 9.9% more likely to be employed, 8.3% more likely to be in the labor force, and yet 62.1% more likely to be in unskilled occupations and

14.5% less likely to be in skilled occupations compared to displaced male owners. Results indicate that the TVA did not favor homeowners over renters, and -in fact- renters are driving the increase in labor market outcomes.

6 Conclusion

Understanding how forced displacement can affect socioeconomic outcomes is an area of concern for researchers and policymakers. We utilize a natural experiment provided by the TVA dam projects to estimate the causal effect of forced displacement on socioeconomic outcomes. We find evidence that forced displacement actually improved individuals labor market outcomes by increasing labor force participation. This is accompanied by a shift from skilled occupations to unskilled occupations.

When organizing displacement, the importance of racial disparities in socioeconomic outcomes and whether government programs contribute to these disparities or ameliorate them should also be of concern. Building on a large body of research showing evidence that government programs can exacerbate racial inequality, we find evidence that the TVA had disparate outcomes for Black and White individuals. Specifically, we find that while Black men had larger increases in labor force participation than White men, they also faced a large increase in unskilled occupation. This can indicate that TVA may have not mitigated the racial disparities between Black and White individuals.

In general, we believe that the TVA dam construction projects had a positive impact on those forced to relocate. To put our estimates in perspective, we use labor estimates from American Community Survey and U.S. Bureau of Labor Statistics data. Applying a 3.3 p.p labor force participation shift to today's average U.S. county workforce of 54,365 would mean instantly adding roughly 1,794 people into the active labor force. A magnitude of 6.4 p.p. applied to today's unskilled labor would transfer 1,922 jobs, while a -4.2 p.p. would be a loss in transfer of roughly 1,022 skilled jobs to the local economy. These estimates suggest an overall positive net benefit in the extensive margin of labor supply but a noticeable effect on the labor heterogeneity.

Our research points to the vital importance of understanding the impacts that reloca-

tion programs have on individuals and how it may vary by race. More research is needed to understand how forced migration can affect where individuals move and whether the specific location of an individual drives the disparities found in our research.

References

- Alderman, D. H. and Brown, R. N. (2010). When a new deal is actually an old deal: The role of TVA in engineering a Jim Crow racialized landscape. In Engineering Earth: The Impacts of Megaengineering Projects. Springer Netherlands.
- Baran, C., Chyn, E., and Stuart, B. A. (2024). The great migration and educational opportunity. American Economic Journal: Applied Economics.
- Becker, S. O. and Ferrara, A. (2019). Consequences of forced migration: A survey of recent findings. Labour Economics.
- Becker, S. O., Grosfeld, I., Grosjean, P., Voigtländer, N., and Zhuravskaya, E. (2020). Forced migration and human capital: Evidence from post-wwii population transfers. American Economic Review.
- Blundell, R., Costa Dias, M., Meghir, C., and Shaw, J. (2016). Female labor supply, human capital, and welfare reform. Econometrica.
- Brown, R. G. (1951). Family removal in the Tennessee Valley. Master's thesis, University of Tennessee.
- Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. Journal of Econometrics, 225(2):200–230.
- Carr, L. J. (1948). Situational Analysis: An Observational Approach to Introductory Sociology. Harper & Brothers.
- Cavallo, E. A., Becerra, O., and Acevedo, L. (2022). The impact of natural disasters on economic growth. In Handbook on the Economics of Disasters. Edward Elgar Publishing.
- Chen, G. (2020). Assessing the financial impact of natural disasters on local governments. Public Budgeting & Finance.
- Chetty, R., Hendren, N., and Katz, L. F. (2016). The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment. American Economic Review.
- Chyn, E. (2018). Moved to opportunity: The long-run effects of public housing demolition on children. American Economic Review.
- Collins, W. J. and Wanamaker, M. H. (2014). Selection and economic gains in the great migration of african americans: new evidence from linked census data. American Economic Journal: Applied Economics.
- Dernoncourt, E. (2022). Can you move to opportunity? evidence from the great migration. American Economic Review.
- Deryugina, T., Kawano, L., and Levitt, S. (2018). The economic impact of hurricane katrina on its victims: Evidence from individual tax returns. American Economic Journal: Applied Economics.
- Duflo, E. and Pande, R. (2007). Dams. The Quarterly Journal of Economics.

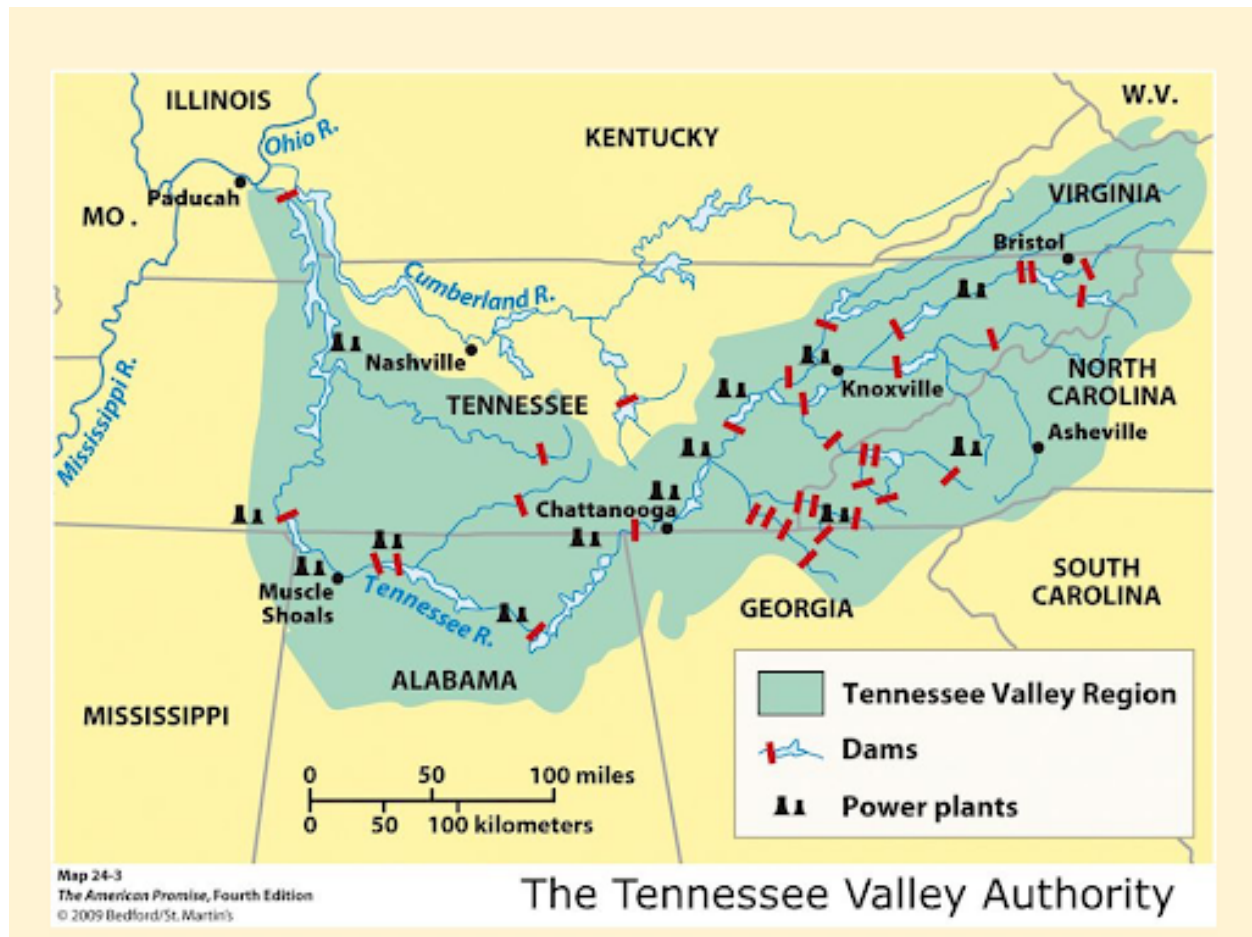
- Eriksson, K. (2019). Moving north and into jail? the great migration and black incarceration. Journal of Economic Behavior Organization.
- Ferrie, J. P. (1997). The entry into the us labor market of antebellum european immigrants, 1840–1860. Explorations in Economic History.
- Fishback, P. and Kantor, S. (2018). New deal studies.
- Fishback, P. V. (2017). How successful was the new deal? the microeconomic impact of new deal spending and lending policies in the 1930s. Journal of Economic Literature.
- Fishback, P. V., Haines, M. R., and Kantor, S. (2007). Births, deaths, and new deal relief during the great depression. The review of economics and statistics.
- Fishback, P. V., Johnson, R. S., and Kantor, S. (2010). Striking at the roots of crime: The impact of welfare spending on crime during the great depression. The Journal of Law and Economics.
- Fishback, P. V. and Kachanovskaya, V. (2016). The politics of the new deal. Technical report, Hoover Institution.
- Fishback, P. V. and Kollmann, T. (2015). Hedonic housing indexes during the great depression. Unpublished manuscript.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. Journal of Econometrics, 225(2):254–277.
- Harrison, A., editor (1991). Black Exodus: The Great Migration from the American South. University Press of Mississippi.
- History.com Editors (n.d.). TVA history - facts, timeline & meaning. <https://history.com>.
- Hornung, E. (2014). Immigration and the diffusion of technology: The huguenot diaspora in prussia. American Economic Review.
- Howard, T. L. (1936). The social scientist in the Tennessee Valley Authority program. Social Forces.
- Johar, M., Johnston, D. W., Shields, M. A., Siminski, P., and Stavrunova, O. (2022). The economic impacts of direct natural disaster exposure. Journal of Economic Behavior Organization.
- Jones, C. M. (2016). Power for the public good: energy, race and class in the United States. PhD thesis, Massachusetts Institute of Technology.
- Karas, D. (2015). Highway to inequity: the disparate impact of the interstate highway system on poor and minority communities in American cities. New Visions for Public Affairs.
- Kelly, M. (2024). Top 10 New Deal programs of the 1930s. <https://www.thoughtco.com/top-new-deal-programs-104687>.
- Kitchens, C. (2013). A dam problem: Tva’s fight against malaria, 1926–1951. The Journal of Economic History.
- Kitchens, C. (2014). The role of publicly provided electricity in economic development: the experience of the tennessee valley authority, 1929–1955. The Journal of Economic History.

- Kline, P. and Moretti, E. (2014). Local economic development, agglomeration economies, and the big push: 100 years of evidence from the tennessee valley authority. The Quarterly Journal of Economics.
- Lim, J., Loveridge, S., Shupp, R., and Skidmore, M. (2017). Double danger in the double wide: Dimensions of poverty, housing quality and tornado impacts. Regional Science and Urban Economics.
- Long, J. and Ferrie, J. (2013). Intergenerational occupational mobility in great britain and the united states since 1850. American Economic Review, 103(4):1109–37.
- McCarthy, C. J. (1949). Land acquisition policies and proceedings in TVA—a study of the role of land acquisition in a regional agency. Ohio State Law Journal.
- Mueller, V. and Quisumbing, A. (2009). Natural disasters and their labor market consequences: evidence from the 1998 flood in Bangladesh. Mimeo, International Food Policy Research Institute (IFPRI).
- Nakamura, E., Sigurdsson, J., and Steinsson, J. (2022). The gift of moving: Intergenerational consequences of a mobility shock. The Review of Economic Studies.
- National Archives and Records Administration (1933). Tennessee valley authority act (1933). <https://archives.gov>.
- National Research Council (2006). Facing Hazards and Disasters: Understanding Human Dimensions. National Academies Press.
- Richardson, E. C. (2022). The race to displace: The long-run causal effect of the 1996 olympic legacy program on residential locations in Atlanta, GA. Master’s thesis, Harvard College.
- Ruggles, S., Flood, S., Sobek, M., Backman, D., Chen, A., Cooper, G., Richards, S., Rogers, R., and Schouweiler, M. (2024a). IPUMS USA: Version 15.0 [dataset]. <https://doi.org/10.18128/D010.V15.0>.
- Ruggles, S., Nelson, M. A., Sobek, M., Fitch, C. A., Goeken, R., Hacker, J. D., Roberts, E., and Warren, J. R. (2024b). IPUMS Ancestry full count data: Version 4.0 [dataset]. <https://doi.org/10.18128/D014.V4.0>.
- Sarvimäki, M., Uusitalo, R., and Jäntti, M. (2022). Habit formation and the misallocation of labor: evidence from forced migrations. Journal of the European Economic Association.
- Satterfield, M. H. (1937). The removal of families from Tennessee Valley Authority reservoir areas. Social Forces.
- Sengupta, R., Coondoo, D., and Routs, B. (2007). Impact of a highway on the socio-economic well-being of rural households living in proximity. Contemporary Issues Ideas in Social Sciences.
- Shin, H. B. and Li, B. (2013). Whose games? the costs of being “olympic citizens” in beijing. Environment and Urbanization.
- Strobl, E. and Strobl, R. O. (2011). The distributional impact of large dams: Evidence from cropland productivity in africa. Journal of development Economics.

- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics, 225(2):175–199.
- Sundstrom, W. A. (2001). Discouraging times: The labor force participation of married black women, 1930–1940. Explorations in Economic History.
- Tennessee Valley Authority (n.d.). Hydroelectric power system. <https://tva.com>.
- Tolnay, S. E. (2003). The african american “great migration” and beyond. Annual Review of Sociology.
- United Nations High Commissioner for Refugees (2026). Figures at a glance. <https://unhcr.org>.
- U.S. Bureau of Labor Statistics (2026). Local area unemployment statistics: County data. <https://bls.gov>. Accessed: 2026.
- U.S. Census Bureau (2024). Table DP03: Selected economic characteristics (Occupation). [https://data.census.gov/table?q=DP03+&t=Occupation&g=010XX00US\\$0500000&y=2024](https://data.census.gov/table?q=DP03+&t=Occupation&g=010XX00US$0500000&y=2024). Accessed: 2026.
- Vigdor, J. L. (2007). The katrina effect: was there a bright side to the evacuation of greater new orleans? The BE Journal of Economic Analysis Policy.
- Walker, M. (1998). African americans and tva reservoir property removal: Race in a new deal program. Agricultural History.

Figures

Figure 1: Map of Tennessee Valley Authority's dam construction projects, 1933-1953



Notes: Map is from *The American Promise, Fourth Edition*.

Figure 2: Sample Relocation Application

WPA 1935
Tennessee Valley Authority
497
Social and Economic Division

Relocation No. 923
Date 8/2/35
Fieldman 82
County 82
Civil District

1. Name
2. P. O. address Athens, Tenn.
3. Highway or road on which home is located
4. Miles from home to nearest elementary school
5. Miles from home to nearest high school
6. Miles from home to nearest church
7. Chief trade center for family
8. Average number of trips there per month
9. Years lived at present residence 24
10. Information concerning houses lived in during 1934 and 1935

Year	Month	Room	Age	Sex	Color	Religion	Education	Occupation
1934	Jan	5	3	35	ma	white	high school	farm hand

11. Source of water supply: Well, Spring, River, Lake, Pond, Cistern, Other
12. Toilet facilities: Flush, Outside, Other
13. Marital status: M, F, S, W, D, R, O, N, P, Q, R, S, T, U, V, W, X, Y, Z
14. Information concerning husband and wife

Name	Age	Sex	Color	Religion	Education	Occupation	Condition of Health
Husband	59	M	white	Methodist	8th	farmer	Good
Wife	57	F	white	Methodist	8th	housewife	Good

15. No. of children born: Boys 5, Girls 5, who have died: Boys 2, Girls 2
16. The newspapers and magazines that the household receives regularly
17. The approximate number of all books in the home 50
18. Educational preferences: Baptist, Membership in other organizations
19. Amount of insurance: Life, Accident and Health, Fire, Marine, Other
20. Liabilities: Mortgage, Other debts, Total

21. Information concerning members of the family who left home to work after 1930 but later returned and now live at home

Name	Relation	Type of Work	Less time of work	Date left home	Date returned	Reason
None						

22. Information concerning deaths in the family during past ten years

Name of Person	Date of Death	Residence at Time of Death	Cause of Death	Age
all deceased				

23. Does the family own a car? If so, what make Ford Model A
24. Check ownership of the following personal possessions: Radio, Sewing machine, Piano, Phonograph, Other articles
25. The approximate value of furniture: Bed, Table, Chair, Sofa, Stool, Other
26. What is husband's calling or trade Farming
27. Information on husband's employment during past 10 years (exclude farming): Dates, Kind of Work, Salary, By whom and Cause of Change in Employment
28. Kind of work preferred
29. Information concerning husband's land (1935): Acreage, Total, Wooded, Pastured, Cultivated, Other, Value, Taxes, Rent, Mortgage, Other

Category	Value	Taxes	Rent	Mortgage	Other
Land	12.0	1.2	2.7	1.0	1.8
Buildings	12.0	1.2	2.0	7.8	1.0
Other					

30. Give location of all land other than home farm, stating whether owned or rented
31. Information concerning acres cultivated (1935): Cereals, 30, Cotton, 27, Soybeans, 10, Small grains, 4, Other crops, 0, Total 71
32. How does crop compare with last year

33. Information concerning type of farming and livestock (1935)

Type of Farming	Value	Value	Value	Value	Value	Value
Corn	34	4.60	1.75	5.125	3.15	65.50

34. Type or types of farming preferred
35. If any land is rented, give name of the owner and to whom it is related to this family
36. Is agreement or lease with the owner written or oral? When was agreement signed?
37. Give information concerning the terms of this agreement for year 1935

Name of Owner	Cash Payment to Owner	Share of Crop	Share of Crop
Landowner			

38. If any land is rented out, give name of the tenant and to whom it is related to this family
39. Is farm owned by a 1934-35 cotton mortgage? If so, to whom name of mortgagee, date of mortgage, and amount of mortgage
40. Give the name of any other person who has been paid for any work done on the farm during 1935
41. Farm expenditures and receipts for 1935

Expenditures	Amount	Receipts	Amount
Livestock and poultry	18	Cash	7.00
Feed	1.50	Other	4.25
Tools and machinery	1.50	Other	1.25
Transportation	1.00	Other	1.00
Medical expenses	1.00	Other	1.00
Other (house repair, etc.)	1.00	Other	1.00
Total expenditures	34.00	Total receipts	12.25

42. Information concerning cash income for twelve months ending Dec. 31, 1935

Source	Amount	Source	Amount
Wages		Children away from home	
Pensions		Children at home	
Dividends		Industrial compensation	
Gifts		Other receipts	
Total cash income		Total cash income	12.25

43. If on relief in 1934, give estimated value of groceries, clothing, fuel, etc. (not included in cash reported) \$1.25
44. How does crop compare with last year

45. Family living: (Products furnished by the farm and used by the family, 1935)

Product	Quantity	Value	Product	Quantity	Value
Wheat, ground	2.50	1.25	Wheat, whole	2.50	1.25
Milk, gallons	2.50	1.25	Butter, pounds	2.50	1.25
Eggs, dozen	2.50	1.25	Chicken, whole	2.50	1.25
Poultry, head	2.50	1.25	Pork, whole	2.50	1.25
Pork, whole	2.50	1.25	Beef, whole	2.50	1.25
Other			Other		
Total			Total		

46. Farm machinery: Tractor, Plow, Harrow, Roller, Cultivator, Mower, Hayrack, Sickle, Grain drill, Seed drill, Other

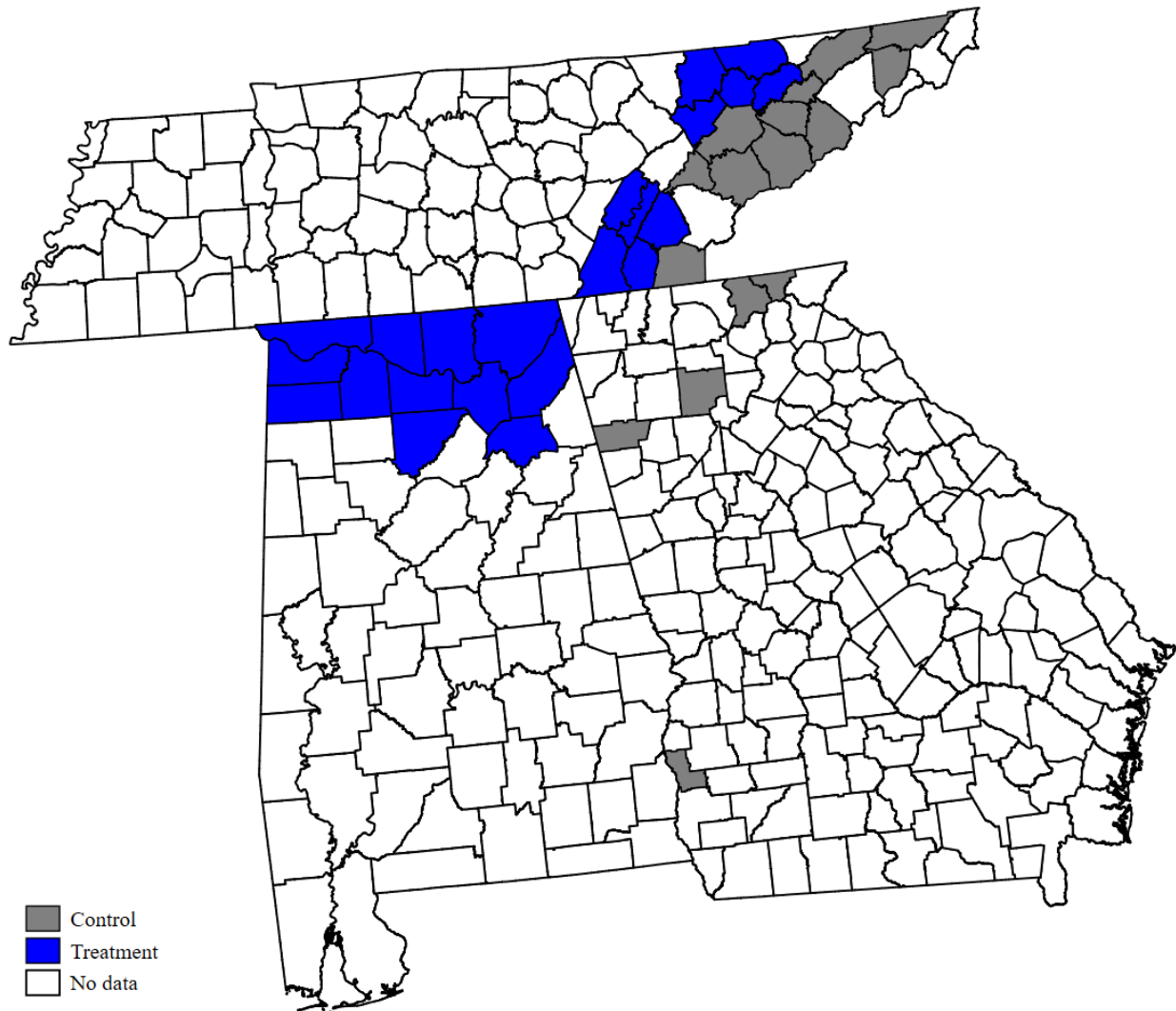
Article	Age	Value	Article	Age	Value
Tractor	2	15	Plow	1	10
Harrow	1	10	Roller	1	10
Cultivator	1	10	Mower	1	10
Hayrack	1	10	Sickle	1	10
Grain drill	1	10	Seed drill	1	10
Other			Other		
Total			Total		

47. The new location has been definitely selected: Yes, No
48. Reasons for selection: Poor soil, Poor water, Poor location
49. If no location has been selected, does the individual desire to relocate: Yes, No
50. If relocation has been selected, does the individual desire to relocate: Yes, No
51. What type of employment is desired: Farm, Part-time farm and industry, Other non-farm or non-industrial
52. Special trades in which husband is skilled, such as automobile repair, carpenter, brick mason, etc.
53. Give names and addresses of close relatives of this family living in this section, and give relationship to head of the family

Name	Address	Relationship

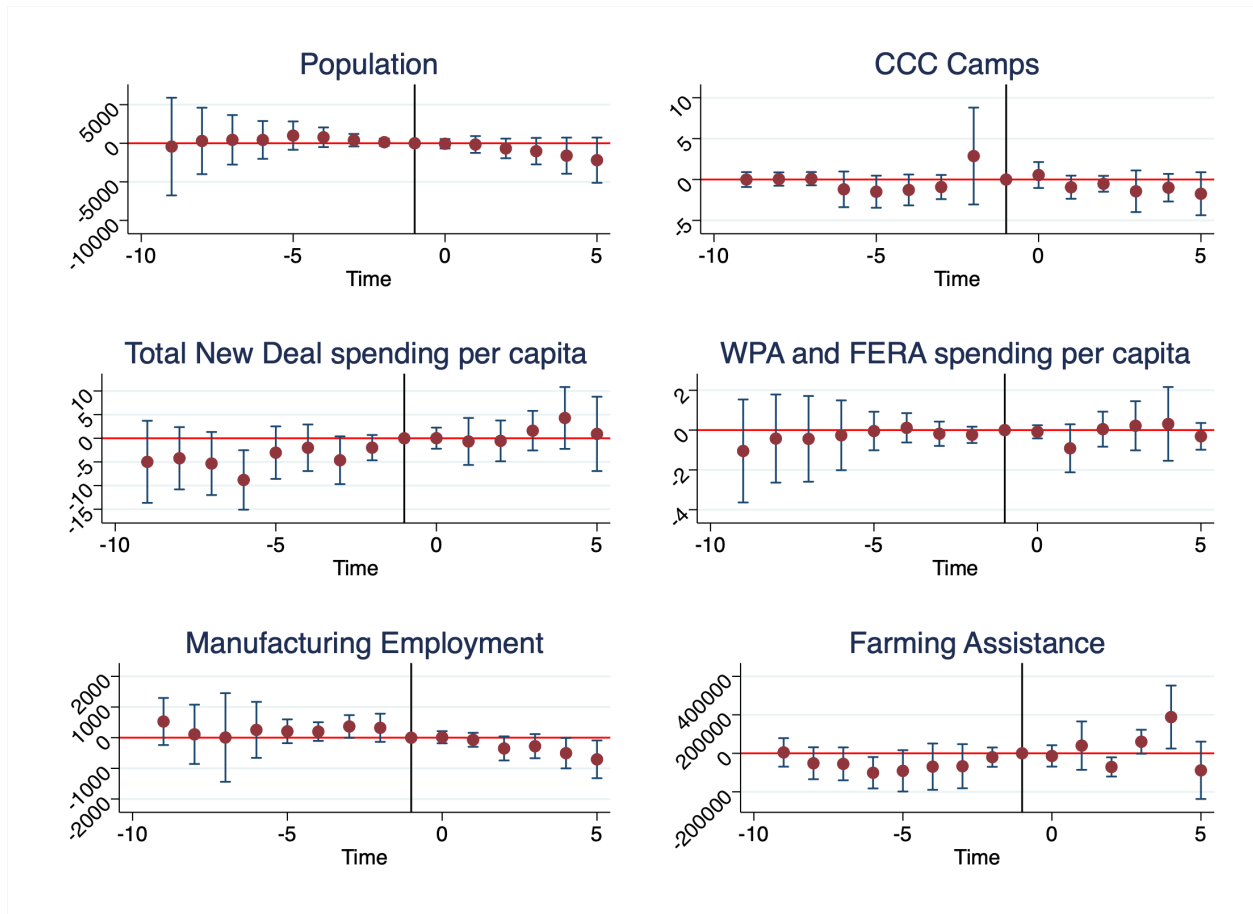
54. How did the individual respond to the interview: Satisfied, Dissatisfied, Indifferent, Interested, Highly Interested
55. What is his attitude toward the TVA: Satisfied, Dissatisfied, Indifferent, Interested, Highly Interested
56. Note any comments regarding special problems or characteristics of this family on the back of last sheet.

Figure 3: Treatment and Control Counties, 1930s
Alabama, Georgia, and Tennessee



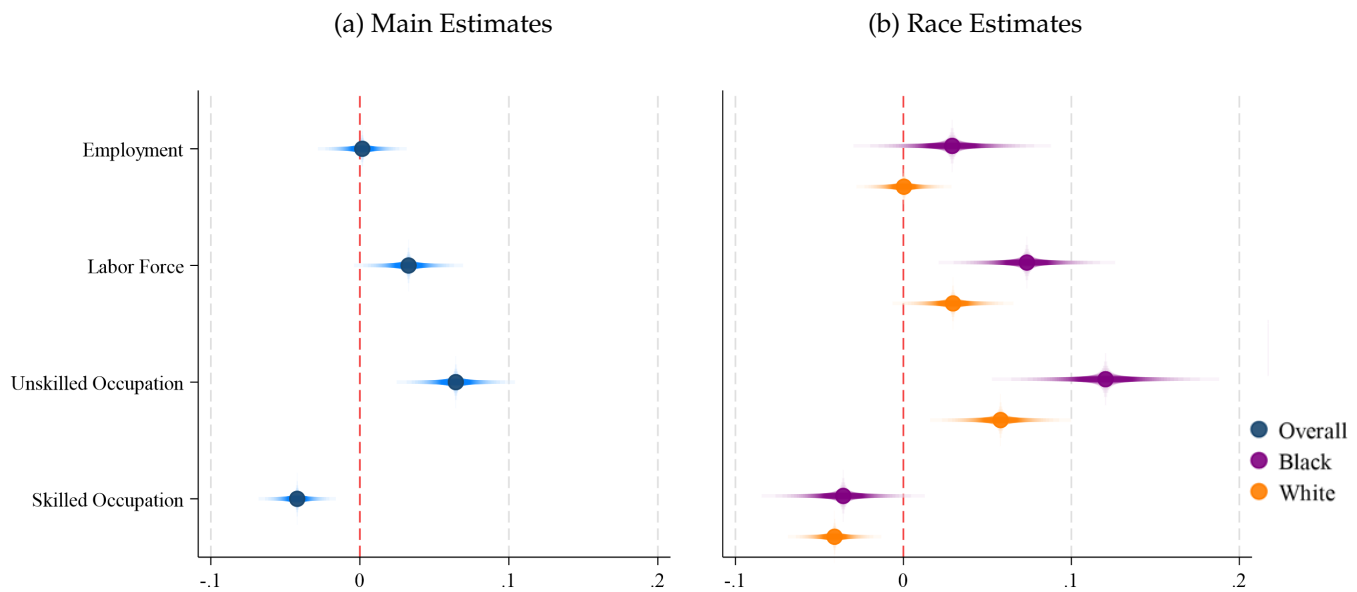
Notes: Our treatment and control groups consists of individuals living in counties that are in three states: Tennessee (top), Alabama (left), and Georgia (right). Counties consisting of treated individuals, prior to displacement, are highlighted in blue, while counties with control individuals are highlighted in gray. Counties without any treated or control individuals are not apart of our sample and are highlighted in white.

Figure 4: Parallel trends tests of annual county-level economic measures



Notes: Figures display event study estimates of the effect of dam construction on county-level proxy outcomes in the years surrounding treatment, as reported in [Fishback and Kantor \(2018\)](#). The sample includes 16 treated and 12 control counties. Each coefficient represents the difference between treated and control counties relative to the year prior to dam construction, which is the omitted period. The vertical line marks the year of dam construction. The red horizontal line is drawn at zero for reference. Confidence intervals are constructed at the 95 percent level and are based on standard errors clustered at the county level. These figures serve as indirect evidence for the parallel trends assumption; because the main analysis relies on Census data, annual pre-treatment trends cannot be assessed directly in the primary sample.

Figure 5: The Average Treatment Effect of a Forced Relocation on Outcomes, Main Results and by Race.



Note: This figure is a point estimate plot of both our (a) main findings and (b) main findings by race. The shaded gradient bands, spreading left and right of our solid-circled point estimates, represent a 95% confidence interval. The vertical dashed red line sits at zero on the x-axis to help visualize statistical significance. The navy blue dots represents our main findings from Table 1 and the orange and purple dots represent our race findings from Table 2, Panels A and B respectively.

Tables

Table 1: Effect of TVA relocation on socioeconomic outcomes

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
Treat*Post	0.002 (0.011)	0.033** (0.013)	0.064*** (0.015)	-0.042*** (0.010)
Observations	317,790	317,790	317,790	317,790
R-squared	0.025	0.037	0.021	0.074
Mean	0.312	0.324	0.119	0.131

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 2: Effect of TVA relocation on socioeconomic outcomes, by race

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.000 (0.010)	0.030** (0.013)	0.058*** (0.015)	-0.041*** (0.010)
Observations	301,251	301,251	301,251	301,251
R-squared	0.026	0.037	0.008	0.073
Mean	0.309	0.321	0.110	0.134
<u>B. Black</u>				
Treat*Post	0.029 (0.022)	0.074*** (0.019)	0.120*** (0.025)	-0.036* (0.018)
Observations	16,454	16,454	16,454	16,454
R-squared	0.020	0.039	0.028	0.022
Mean	0.371	0.395	0.290	0.0722
Coefficients Different By Race	No	Yes	Yes	No

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for White individuals in the main sample, Panel B reports outcomes for Black individuals in the main sample. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 3: Placebo test of the effect of TVA relocation on socioeconomic outcomes

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
Treat*Post	0.022 (0.019)	0.003 (0.015)	-0.001 (0.014)	-0.118*** (0.017)
Observations	164,704	201,908	201,908	201,908
R-squared	0.047	0.068	0.036	0.099
Mean	0.314	0.311	0.108	0.109

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is a placebo indicator (1930) for being after dam construction. Employment Rate has a smaller sample size due to the sample collection of this variable in the 1920's Census. We cluster standard errors at the 1920 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4: Effect of labor market outcomes from 1930 on the probability of being in the treatment group

Variable	Treatment (Std. Err.)
Employment Rate	-0.000 (0.000)
Labor Force Participation	-0.000 (0.000)
Unskilled Occupation	-0.000 (0.000)
Skilled Occupation	0.000 (0.000)
Observations	158,973
R-squared	0.962

Note: We regress treatment status on each outcome in the year 1930. Race controls are included in each regression. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5: Effect of TVA relocation on socioeconomic outcomes, county matched sample

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. All Counties</u>				
Treat*Post	0.021* (0.011)	0.028** (0.011)	0.055*** (0.014)	−0.027*** (0.010)
Observations	418,072	418,072	418,072	418,072
R-squared	0.022	0.039	0.026	0.080
Mean	0.499	0.494	0.177	0.159
<u>B. Exclude Counties with Dams Built in the 1930s</u>				
Treat*Post	0.015 (0.010)	0.023** (0.010)	0.056*** (0.013)	−0.034*** (0.010)
Observations	375,814	375,814	375,814	375,814
R-squared	0.025	0.040	0.028	0.077
Mean	0.498	0.497	0.169	0.172
<u>C. Exclude Counties with Dams Built in the 1940s</u>				
Treat*Post	0.020* (0.011)	0.027** (0.011)	0.058*** (0.014)	−0.022** (0.009)
Observations	423,835	423,835	423,835	423,835
R-squared	0.022	0.039	0.027	0.076
Mean	0.494	0.491	0.179	0.162

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A includes the full county matched sample. Panel B excludes counties where dams were built in the 1930s, and Panel C excludes counties where dams were built in the 1940s. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 6: Effect of TVA relocation on baseline county characteristics in 1930

VARIABLES	(1) 1930 Average Emp. Rate	(2) 1930 Average Labor Force	(3) 1930 Average Unskilled	(4) 1930 Average Skilled
Treat*Post	−0.003 (0.002)	−0.002 (0.004)	0.001 (0.003)	0.006 (0.009)
Observations	317,790	317,790	317,790	317,790
R-squared	0.579	0.707	0.588	0.762
Mean	0.521	0.546	0.200	0.206

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7: Effect of TVA relocation on socioeconomic outcomes, race matched sample

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.010 (0.011)	0.016 (0.012)	0.003 (0.016)	-0.027** (0.012)
Observations	28,181	28,181	28,181	28,181
R-squared	0.029	0.047	0.025	0.054
Mean	0.397	0.382	0.129	0.0626
<u>B. Black</u>				
Treat*Post	-0.028 (0.025)	0.021 (0.021)	0.030 (0.029)	-0.078*** (0.015)
Observations	2,917	2,917	2,917	2,917
R-squared	0.043	0.060	0.062	0.068
Mean	0.399	0.381	0.194	0.0154
Coefficients Different By Race	No	No	No	Yes

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A includes White individuals in the treatment group matched to White individuals in the control group. Panel B includes Black individuals in the treatment group matched to Black individuals in the control group. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8: Effect of TVA relocation on socioeconomic outcomes, by homeownership status

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. Renter</u>				
Treat*Post	0.007 (0.014)	0.035* (0.017)	0.087*** (0.015)	-0.054*** (0.009)
Observations	150,829	150,829	150,829	150,829
R-squared	0.035	0.052	0.025	0.076
Mean	0.298	0.314	0.131	0.123
<u>B. Owner</u>				
Treat*Post	-0.030** (0.014)	0.003 (0.013)	0.014 (0.015)	-0.036*** (0.013)
Observations	166,961	166,961	166,961	166,961
R-squared	0.017	0.025	0.019	0.074
Mean	0.327	0.335	0.108	0.139
Coefficients Different By Owner	Yes	Yes	Yes	Yes

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for renters in the main sample, Panel B reports outcomes for homeowners in the main sample. Race controls are included in each regression. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

7 Appendix

Figure A1: Stages of dam construction mentioned by Howard (1936)

- ▶ Step 1: Both engineers and social scientists must make numerous investigations. The engineers want to find a firm, physical foundation for their structure and for the finished product to withstand the forces of nature. Before the project is approved or the site is determined, the social scientists need to figure out if the anticipated benefits outweigh the monetary and social costs of the enterprise. If so, they should proceed with the dam construction.
- ▶ Step 2: After the decision to build the dam has been made, there should be focus on questions relating to construction. What labor supply is available within a reasonable distance of the project? What is the composition of the population? What skills are represented by the population? What race factors are present among the population?
- ▶ Step 3: There should be consideration of the need for a camp to house employees after they are hired which includes awareness of all social implications involved in housing workers in construction communities.
- ▶ Step 4: Understanding the people who are to be employed is required. What skills are needed? What type of employee training will be most useful? Will the training be on the construction site or will it be held at workers homes?
- ▶ Step 5: Studies are to be made to assist in planning the educational facilities needed by the employees of the Authority.
- ▶ Step 6: As the actual construction begins, there are a new series of questions to be considered. What commercial facilities must be provided? Are there any desirable recreational facilities? What other economic and social needs must be anticipated?
- ▶ Step 7: Determine the type of community organization that will be most effective such as the community making decisions within popular participation.
- ▶ Step 8: Figure out what cooperative arrangement can be worked out that will be mutually satisfactory to the Authority and the local governments. An example is to consider a reexamination of health facilities due to new demands.
- ▶ Step 9: Social scientists must determine detailed facts concerning families to be displaced by the reservoir.
- ▶ Step 10: As reservoir land is acquired and the people move out of the region, more questions arise for the public administration and economist. What will be the financial status of the local government after a large portion of an area has been purchased? What source of revenue is left? What is the debt and can it be paid? What is the future of this particular part of the local government?
- ▶ Step 11: As the construction project nears completion, new problems arise. What vocational opportunities are open to workers whose period of employment is about to end? Can the Authority assist them in their efforts to utilize the experience and training received during this period of employment?
- ▶ Step 12: Finally, important adjustments must be made after the construction project is over and people who were involved begin to leave the region.

Table A1: Summary statistics, treatment and control groups

	TG	CG	Difference
<i>Dependent Variables</i>			
Employment	0.397 [0.489]	0.508 [0.500]	-0.112*** (0.011)
Labor Force	0.409 [0.492]	0.542 [0.498]	-0.134*** (0.015)
Unskilled Jobs	0.135 [0.341]	0.187 [0.390]	-0.053*** (0.011)
Skilled Jobs	0.082 [0.275]	0.222 [0.415]	-0.139*** (0.048)
<i>Independent Variables</i>			
White	0.906 [0.292]	0.952 [0.213]	-0.046* (0.026)
Black	0.094 [0.292]	0.048 [0.213]	0.046* (0.026)
Observations	7,924	150,674	

TG refers to individuals in treated counties and CG refers to individuals in control counties. Means are reported for each group with standard deviations in brackets. The Difference column reports coefficients from OLS regressions of each outcome on treatment. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Linked vs. Unlinked Summary Statistics

	White	Male	Rented	Farmers	Total Individuals
<u>Panel A: Linked to 1930 Census</u>					
Linked	88.79%	56.53%	62.29%	71.04%	18,933
Not linked	77.90%	57.75%	77.72%	67.64%	5,460
<u>Panel B: Linked to 1940 Census</u>					
Linked	90.24%	57.38%	62.35%	73.81%	15,443
Not linked	82.33%	52.77%	62.00%	58.76%	3,482

This table reports summary statistics for individuals linked and not linked to the 1930 and 1940 censuses, pooling residents of Alabama, Tennessee, and Georgia who were displaced by TVA dam construction. Panel A compares individuals who were and were not successfully linked from the treatment sample to the 1930 census. Panel B restricts to individuals successfully linked to the 1930 census (Panel A, Linked row) and compares those who were and were not successfully linked to the 1940 census. White, Male, Rented, and Farmers report the share of each group with the corresponding characteristic as recorded in the linked census year. Total Individuals reports the number of observations in each row.

Table A3: Unskilled Occupations Breakdown, Overall and by Race

VARIABLES	(1) Unskilled (Overall)	(2) Unskilled Without Farming	(3) Unskilled Farming
<u>A. Overall</u>			
Treat*Post	0.064*** (0.015)	0.014** (0.006)	0.009** (0.004)
Observations	317,790	317,790	317,790
R-squared	0.021	0.031	0.040
Mean	0.119	0.064	0.150
<u>B. Black</u>			
Treat*Post	0.120*** (0.025)	0.009 (0.016)	0.007 (0.007)
Observations	16,454	16,454	16,454
R-squared	0.028	0.051	0.039
Mean	0.290	0.214	0.137
<u>C. White</u>			
Treat*Post	0.058*** (0.015)	0.012** (0.006)	0.009* (0.005)
Observations	301,251	301,251	301,251
R-squared	0.008	0.011	0.040
Mean	0.110	0.056	0.151
Coefficients Different By Race	Yes	No	No

Unskilled (Overall) is from our main results and is here for comparison. Unskilled Without Farming considers our unskilled occupations from our main results, but codes any unskilled farming occupations as zero. Unskilled Farming codes any related unskilled farming occupation as one and the rest as zeros. Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A is the overall estimates for each classification of unskilled occupations. Panel B estimates each classification of unskilled occupations for Black individuals while Panel C does so for White individuals. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A4: Effect of TVA relocation on socioeconomic outcomes, county matched sample by race

VARIABLES	(1) Employment Rate	(2) Labor Force Participation	(3) Unskilled Occupation	(4) Skilled Occupation
<u>A. White</u>				
Treat*Post	0.018* (0.011)	0.025** (0.011)	0.052*** (0.015)	-0.031*** (0.011)
Observations	374,775	374,775	374,775	374,775
R-squared	0.023	0.039	0.012	0.085
Mean	0.500	0.494	0.163	0.166
<u>B. Black</u>				
Treat*Post	0.037* (0.022)	0.049** (0.018)	0.090*** (0.024)	-0.004 (0.012)
Observations	43,167	43,167	43,167	43,167
R-squared	0.015	0.036	0.041	0.067
Mean	0.486	0.491	0.299	0.103
Coefficients Different By Race	No	No	No	No

Year and county fixed effects are included in each regression. Treat is an indicator for being displaced by TVA dam construction. Post is an indicator for being after dam construction. Panel A reports outcomes for White individuals in the main sample, Panel B reports outcomes for Black individuals in the main sample. Both panels include the full county matched sample. We cluster standard errors at the 1930 county level. Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$